

Bad Face Detection

Code is available in landmarks.ipynb. I used Jupyter and Voxel51 because visual exploration is useful during early experimentation and testing.

Four facial-position features were calculated as described in face_position_calculation.md.

Defining Bad Face

- In a real ML project, the key step is to define BAD FACE: the exact thresholds and business rules.
- Check whether thresholds should differ by age, gender, or race.
- Decide how to handle face distortions such as a broken nose.
- Choose thresholds based on the downstream task and measure which face positions make the next model fail.

If bad-face detection is a preprocessing step for classification, I would use the same landmark model on the classification dataset and analyze when the classifier succeeds or fails.

With more time, I would validate this using a second landmark detector.

Face Classification

Exploration and Cleaning

- The dataset contained many duplicates and near-duplicates. I cleaned them using hash checks, voxel uniqueness, and CLIP feature-vector similarity.
- Some images had no visible face, some were grayscale, and many were very blurry.
- A t-SNE visualization showed mislabeled images. I did not have enough time to fully clean those labels.
- The age distribution was highly imbalanced, which I did not handle in this iteration.

Experiments

All training was tracked with MLflow.

- Tested multiple loss functions: UncertaintyMultiTaskLoss, Simplified UncertaintyMultiTaskLoss, manually weighted combined cross-entropy, and combined cross-entropy without weights.
- Compared different dataset versions: uncleaned and cleaned.

Results

Results for UncertaintyMultiTaskLoss with the cleaned dataset:

Confusion matrix - gender

confusion_matrices/test_best_model_gender_confusion_matrix.csv 30B



Path: file:///C:/Users/anspa/au10tix/classification/mlruns/1/408ae73c686e4d019952c323febd8705/artifacts/confusion_matrices/test_best_model_gender_confusion_matrix.csv

Previewing the first 2 rows

	0	1
0	1556	139
1	153	1362

Confusion matrix - race

0	1	2	3	4
1041	45	42	86	145
12	563	16	21	20
25	6	378	19	28
29	59	13	391	49
36	31	23	39	93

- Validation gender accuracy: 0.9269
- Validation race accuracy: 0.8077
- Validation race balanced accuracy: 0.7442
- Per-class metrics were tracked as well.

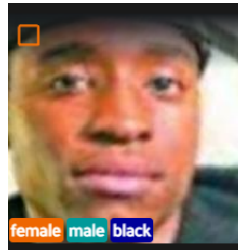
Further Experiments

- Clean images more aggressively by excluding blurry, bad-pose, and grayscale images.
- Perform deeper results analysis and use it to drive the next development iterations.
- Try training in two stages instead of training both tasks simultaneously.
- Simplify and clean the code further.

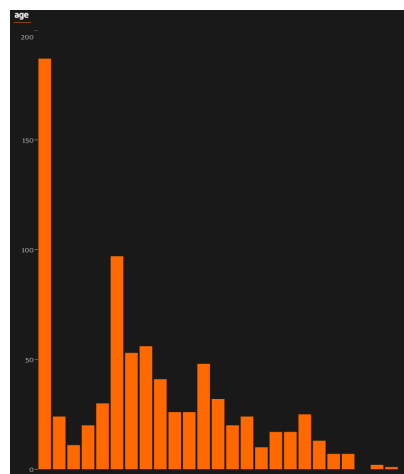
Results Analysis

Gender

Correct prediction, but mislabeled example

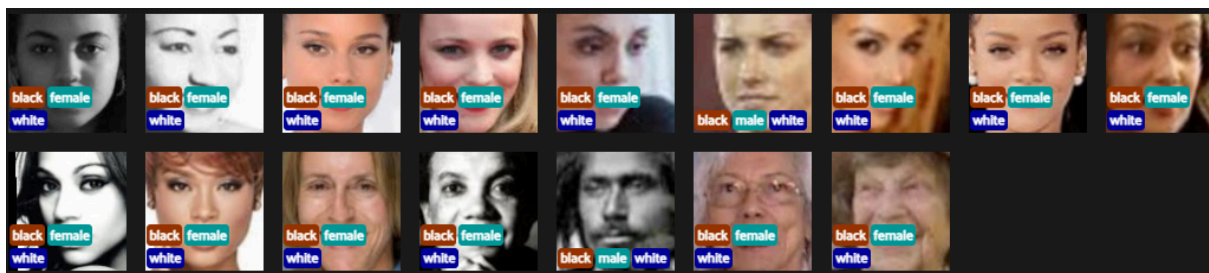


Predicted male, labeled female. Many gender errors appear to involve babies.

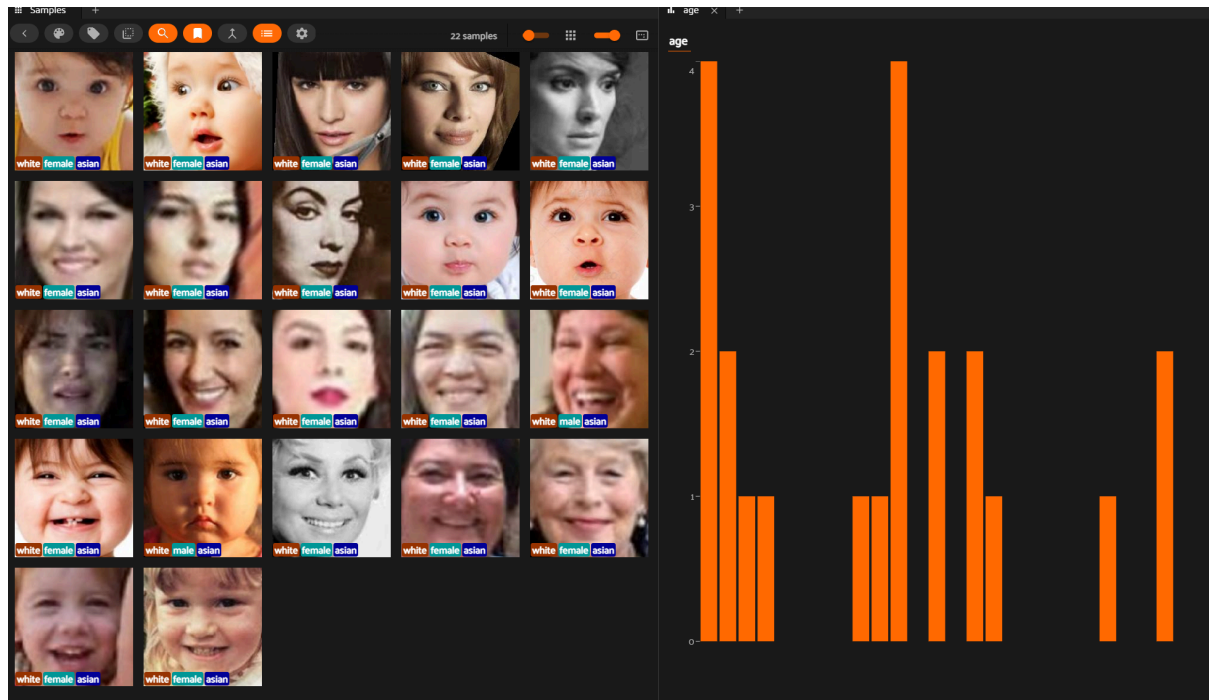


Race

Predicted white, labeled black. Only a few were misclassified; most examples were grayscale.



Predicted white, labeled Asian. The model may struggle more with babies, labels are not always unambiguous, and squeezed eyes appear to affect predictions.



Production

Top 3 production changes for 1,000 faces/sec:

1. Optimized inference service: batching, GPU/CPU autoscaling, ONNX/TensorRT if useful, and low-latency APIs instead of notebook/script prediction.
2. Monitoring and quality checks: latency, throughput, errors, confidence, data drift, and prediction distribution.
3. Reliable scalable pipeline: queues or streams, backpressure, retries, timeouts, versioned models, and safe fallback behavior.